DO NOW — RECALL

Karnaugh maps 2: simplify to gates

Name: _____ Class: _____ Date: _____

1. In a Karnaugh-map pair, A stays 0 and B changes between 0 and 1. What is the simplified term?

2. Show the array after one complete left-to-right pass of ascending bubble sort on [8, 3, 6, 2] .

3. C#: int[,,] scores = new int[2, 3, 4]; How many values can it store? What does scores.GetLength(2) return?

4. Add 01000000 0011 and 01000000 0010 . Format: 8-bit two’s-complement fractional mantissa + 4-bit two’s-complement exponent . Show exponent alignment and the final normalised result.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

¬A A stays 0, so retain NOT A (¬A). B changes, so remove B.

2. Model answer

[3, 6, 2, 8] Swap 8 and 3 → [3, 8, 6, 2]; swap 8 and 6 → [3, 6, 8, 2]; swap 8 and 2 → [3, 6, 2, 8]. The largest value is now last.

3. Model answer

24 values; GetLength(2) = 4 2 layers × 3 rows × 4 columns = 24 values. Dimension 2 is the third dimension (columns).

4. Model answer

01100000 0011 (denary 6) Align to exponent 3 by shifting the second mantissa right: 01000000 → 00100000. Add the mantissas: 01000000 + 00100000 = 01100000. The exponent remains 0011. This is normalised: $0.75 \times 2^3 = 6$.

By the end, you can…

- choose efficient K-map groups including wraparound and overlap
- derive a simplified expression from each group
- translate the simplified expression into an equivalent gate plan

Vocabulary — know these terms

Recall the term — efficient group

Recall the term — overlap

Recall the term — wraparound

Recall the term — literal

Recall the term — product term

Recall the term — simplified expression

Recall the term — equivalent

Recall the term — gate plan

NEW LEARNING

Read, annotate and explain

Start with the map

Find the simplest rule for these outputs.

Method: Circle → Compare → Remove changing inputs → Write expression → Draw gates.

Find the 1s

Look at the 2×2 map on the left. Three squares contain 1 . We need to cover them in groups of neighbouring 1s.

Use rectangles containing 1, 2, 4, 8 or 16 cells. Include only 1s, and choose the largest useful groups. Diagonal cells are not neighbours.

A square may belong to two groups

This is overlap . It is allowed when it makes the expression simpler.

Keep the inputs that stay the same

Bottom pair: A stays 1; B changes. Keep A.

Right pair: B stays 1; A changes. Keep B.

The outside columns form one group

BC runs 00, 01, 11, 10 . The first and last columns are neighbours: only B changes between 00 and 10 .

Imagine rolling the map so its left and right edges meet. The four 1s make a 2 × 2 group ; the middle 0s are outside it.

Compare all four cells

A changes between the two rows → remove A.

B changes between columns 00 and 10 → remove B.

C stays 0 in all four cells → keep NOT C.

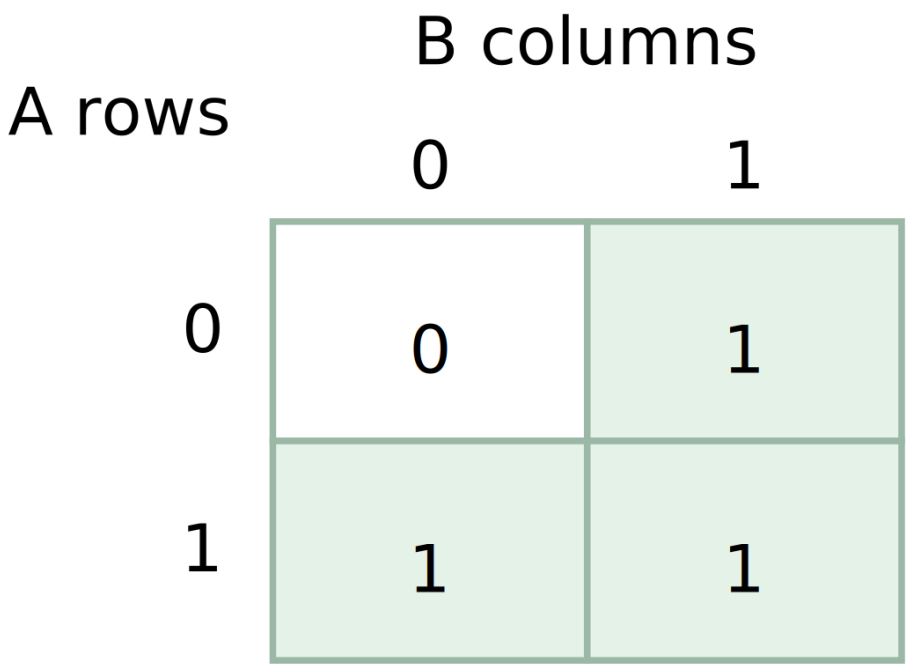
Gate plan: C → NOT gate → Q.

Explain one step in your own words, using the correct technical terms.

NEW LEARNING · original source diagrams

Start with the map

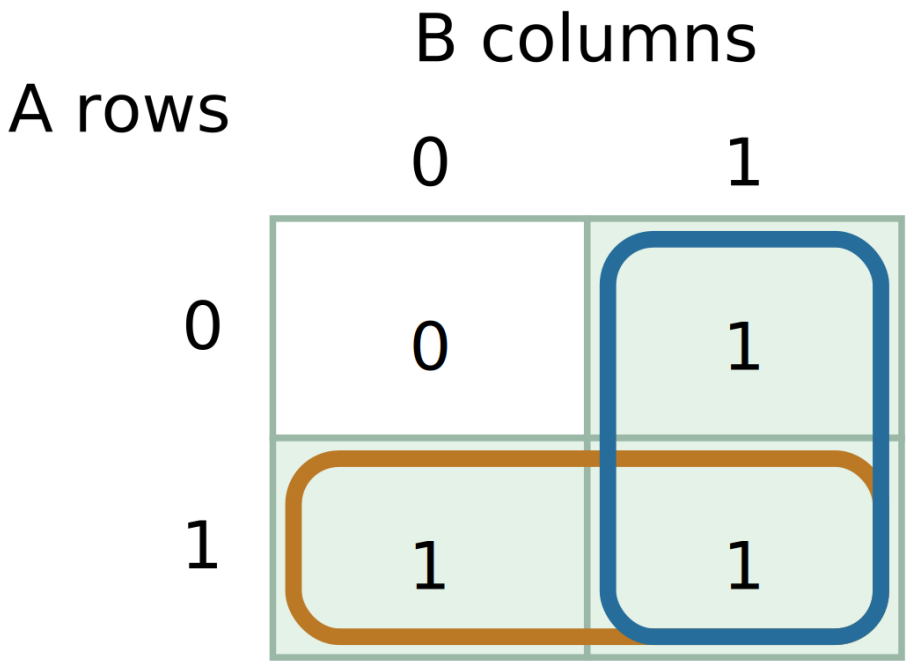
Find the simplest rule for these outputs.



A rows and B columns Karnaugh map

A square may belong to two groups

This is overlap . It is allowed when it makes the expression simpler.

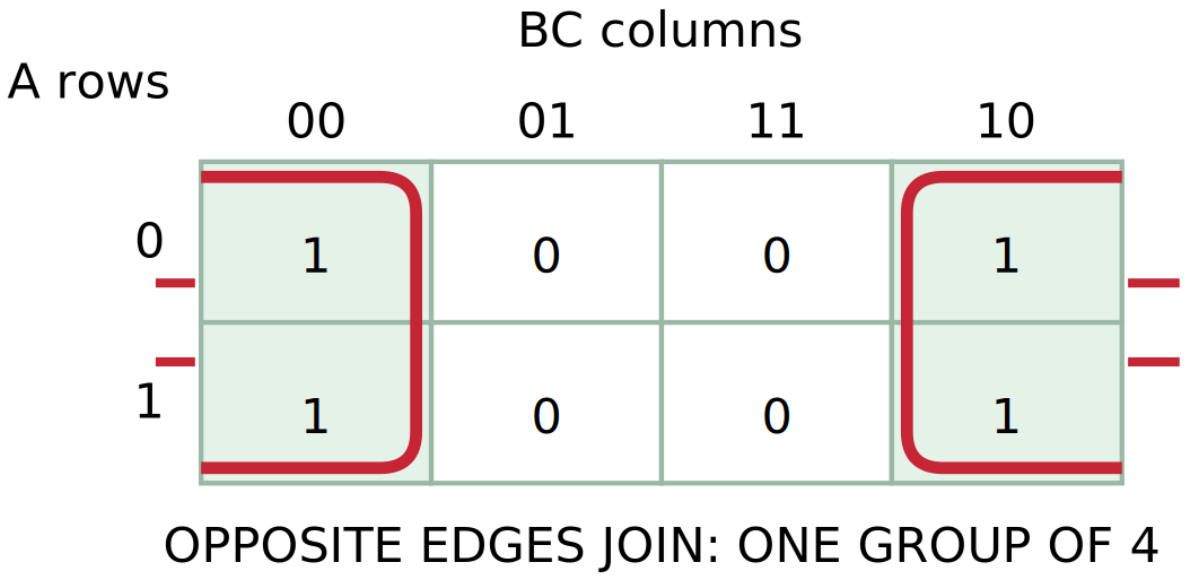


A rows and B columns Karnaugh map

NEW LEARNING · original source diagrams

The outside columns form one group

BC runs 00, 01, 11, 10 . The first and last columns are neighbours: only B changes between 00 and 10 .



The red outline continues across the left and right edges. All four 1s belong to one group.

I DO — WORKED EXAMPLE

Follow the worked method

How do two groups become a circuit?

Find the two pairs of 1s

ORANGE · TOP: circle columns 01 and 11 where A = 0.

BLUE · BOTTOM: circle columns 11 and 10 where A = 1.

Follow the same coloured groups on the map on the left. Each pair becomes one term.

Keep ¬A and C

A = 0 in both squares, so write ¬A . C = 1 in both, so keep C . B changes , so remove B.

Keep A and B

A = 1 in both squares, so keep A . B = 1 in both, so keep B . C changes , so remove C.

Join the two pairs with OR

Either coloured pair makes Q = 1 . AND the inputs inside each pair; OR the two results together.

Draw the actual logic gates

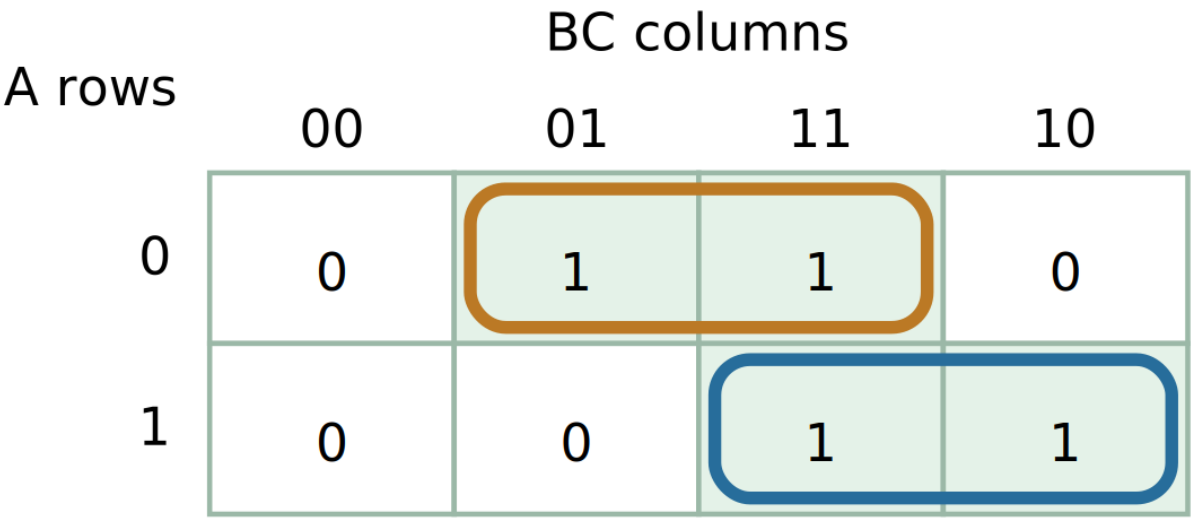
Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

How do two groups become a circuit?

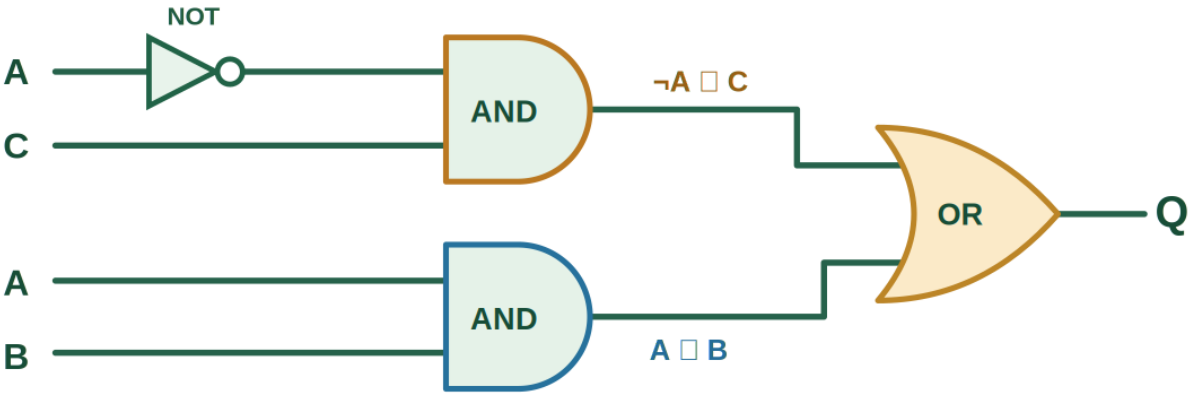
Rows A; columns BC in Gray-code order.



A rows and BC columns Karnaugh map

Draw the actual logic gates

Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.



Logic circuit: input A passes through a NOT gate to the top AND with C; A also goes directly to the lower AND with B. The two AND outputs feed an OR gate, producing Q.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Find one pair

Method / working:

2. Let two groups share a 1

Method / working:

3. Remember the outside edges touch

Method / working:

4. Try a bigger Karnaugh map

Method / working:

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find one pair

Circle the two 1s. Which two inputs always equal 1? Which input changes?

A rows \ BC columns	00	01	11	10
0	0	0	0	0
1	0	1	1	0

A rows and BC columns Karnaugh map

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Find one pair 1. Find the 1s Circle the two 1s. Which two inputs always equal 1? Which input changes?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Let two groups share a 1

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 3: Remember the outside edges touch

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 4: Try a bigger Karnaugh map

DISCUSS — Compare intermediate values and your reasoning. Justify one step using the relevant rule, representation or truth table.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

1. Find one pair
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

Question 2

2. Let two groups share a 1
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

3. Wrap around the outside edges
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

Question 4

4. Two groups in a bigger map
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression P.
Draw the logic gates and connections.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A K-map simplification is valid only if the simplified expression gives the same output for every possible input combination.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

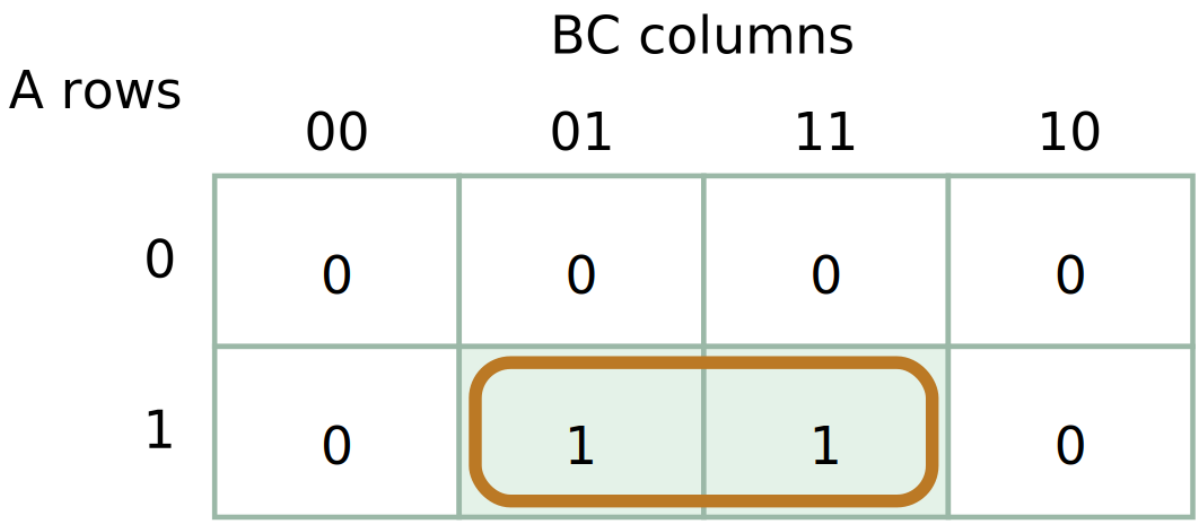
Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO CHECK · original source diagrams

2. Compare your circles

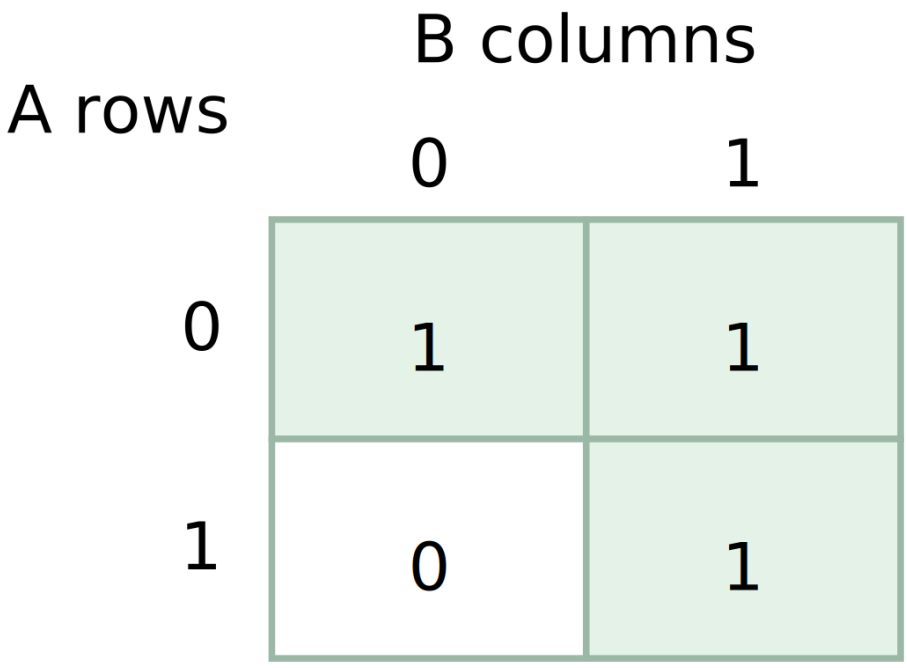
The pair is on row A=1, across columns BC=01 and 11.



A rows and BC columns Karnaugh map

Let two groups share a 1

Find the two largest pairs. Is it okay for the top-right 1 to be used twice?

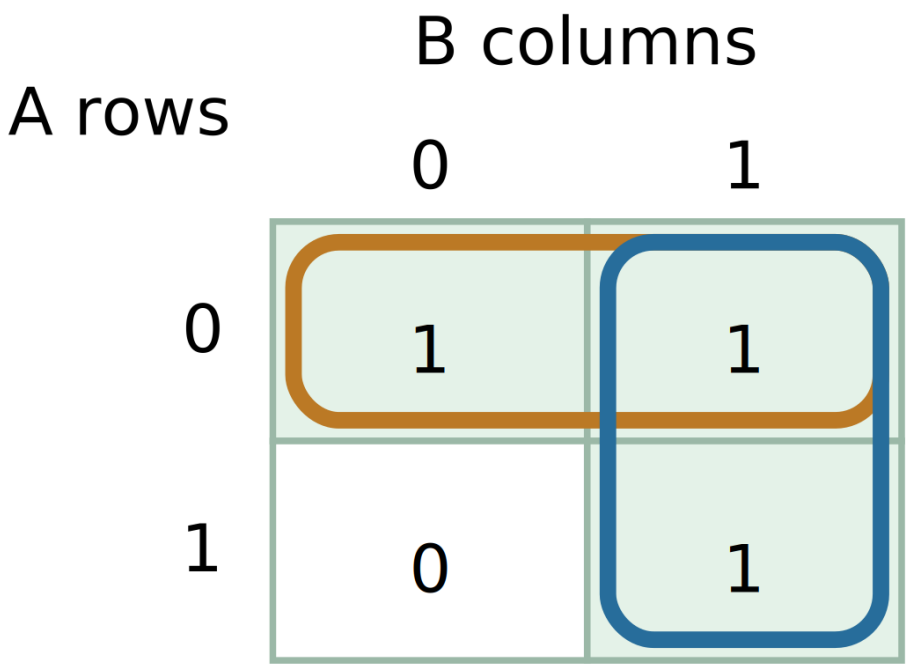


A rows and B columns Karnaugh map

WE DO CHECK · original source diagrams

2. Compare your circles

Top row and right column both contain a valid pair, sharing the top-right 1.



A rows and B columns Karnaugh map

Remember the outside edges touch

The BC columns are 00, 01, 11, 10. Circle the four 1s at the two outside edges as ONE group.

BC columns

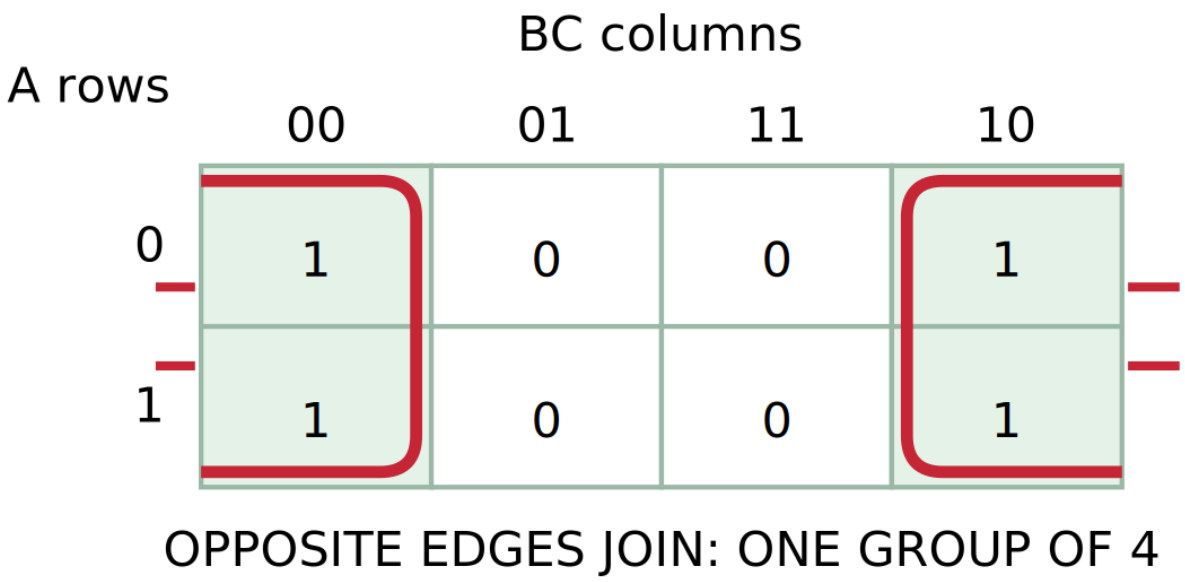
A rows	00	01	11	10
0	1	0	0	1
1	1	0	0	1

A rows and BC columns Karnaugh map

WE DO CHECK · original source diagrams

2. Compare your circles

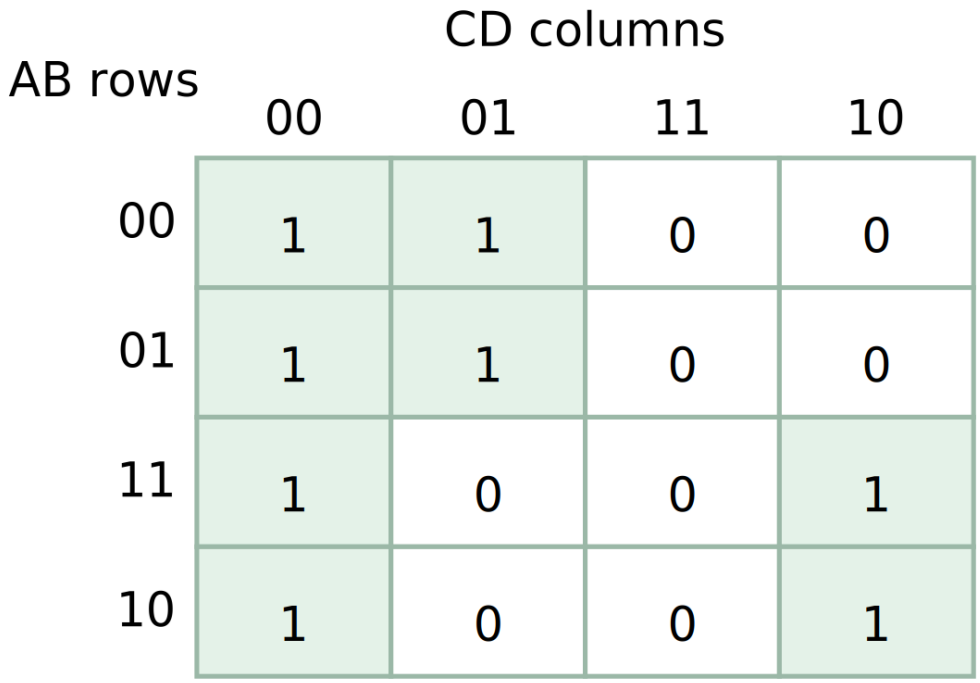
The outside columns (BC=00 and BC=10) are neighbours across the edge.



One wraparound group of four: Q equals NOT C

Try a bigger Karnaugh map

Rows AB and columns CD use Gray code. Find two groups of four: top-left, then bottom-left and bottom-right across the edges.

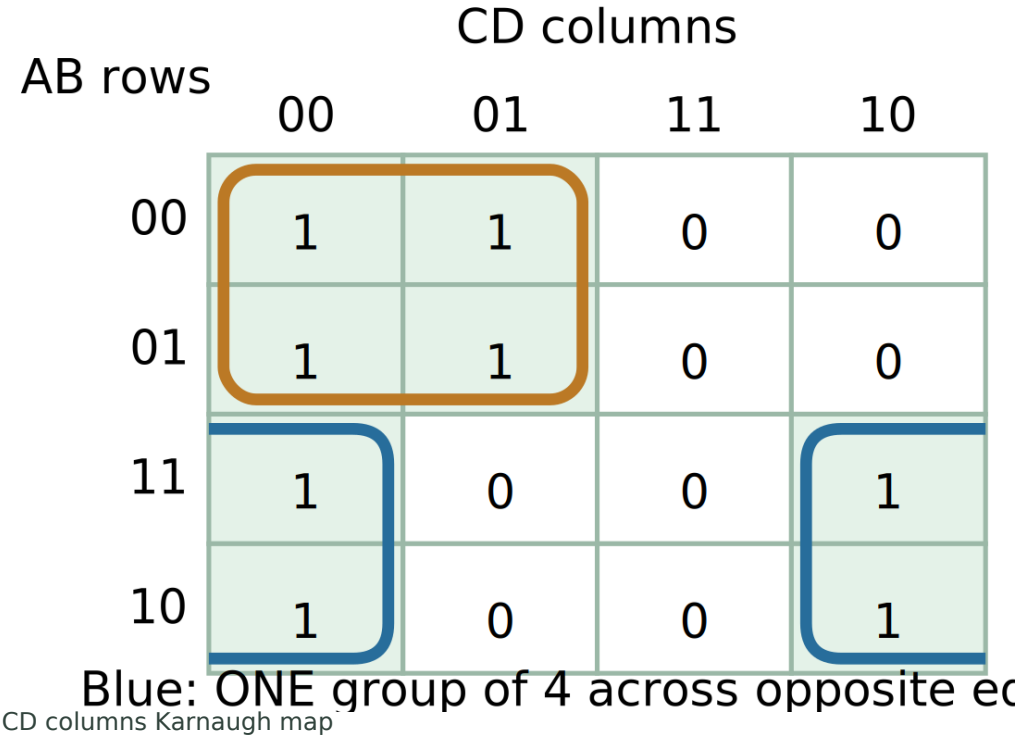


AB rows and CD columns Karnaugh map

WE DO CHECK · original source diagrams

2. Compare your circles

First group: top-left 2×2. Second group: the first and last columns in the bottom two rows form one wraparound group of four.



AB rows and CD columns Karnaugh map

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

2. Compare your circles

The pair is on row A=1, across columns BC=01 and 11.

3. Simplify the group(s)

Use the two unchanged inputs to make one Boolean term. Which gate joins them?

4. Why is this correct?

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

1. Find the 1s

Find the two largest pairs. Is it okay for the top-right 1 to be used twice?

2. Compare your circles

Top row and right column both contain a valid pair, sharing the top-right 1.

3. Simplify the group(s)

Top pair: A stays 0. Right pair: B stays 1. Write the two terms and join them.

4. Why is this correct?

IMPROVE: Explain one correction to your own reasoning.

Guided 3 — original CHECK

1. Find the 1s

The BC columns are 00, 01, 11, 10. Circle the four 1s at the two outside edges as ONE group.

2. Compare your circles

The outside columns (BC=00 and BC=10) are neighbours across the edge.

3. Simplify the group(s)

Across the four 1s, which variable stays 0? Which two change? Write Q.

4. Why is this correct?

IMPROVE: Explain one correction to your own reasoning.

Guided 4 — original CHECK

1. Find the 1s

Rows AB and columns CD use Gray code. Find two groups of four: top-left, then bottom-left and bottom-right across the edges.

2. Compare your circles

First group: top-left 2×2. Second group: the first and last columns in the bottom two rows form one wraparound group of four.

3. Simplify the group(s)

Write the fixed inputs for each group. Make each term, join with OR, and plan the gates.

4. Why is this correct?

IMPROVE: Explain one correction to your own reasoning.